

Decision-Chain Alignment: Method Comparison

Steering a frozen 12L-8H-512D GPT-NeoX toward decision function f

Setup

A pretrained model picks decision function f vs. g at each chain step $\approx 50/50$. Each method tries to push the model toward always picking f . Two evaluation slices:

- **val_full** — mixed-pattern val (the honest test for off-distribution alignment).
- **val_ffff** — all- f val (in-distribution sanity check).

Metrics: **f-sel** = per-step fraction picking f 's letter %; **Full align** = strict per-example all- f %; **Op-acc** = arithmetic correctness of generated blocks %.

Table 1: **val_full** (mixed-pattern). Δ = change vs. baseline in percentage points. *Memory alignment matches or beats every finetuning baseline on alignment while preserving arithmetic best.*

Method	f-sel (%)	Full align (%)	Op-acc (%)
Baseline (no training)	52.5	8.8	94.8
GRPO (per-step f reward) [†]	60.2 +7.7	11.4 +2.6	93.9 -0.9
LoRA, mode A	81.6 +29.1	37.2 +28.4	94.3 -0.5
LoRA, mode B	96.6 +44.1	83.5 +74.6	89.3 -5.5
Naive FT (full), mode A	96.0 +43.5	84.2 +75.3	92.6 -2.1
Naive FT (full), mode B	92.9 +40.5	69.8 +60.9	84.7 -10.0
Memory align (L=2, N=64)	97.6 +45.1	88.0 +79.2	94.1 -0.7

Table 2: **val_ffff** (all- f , in-distribution sanity check). Every aligning method saturates near the ceiling.

Method	f-sel (%)	Full align (%)	Op-acc (%)
Baseline (no training)	76.8	64.8	95.1
GRPO (per-step f reward) [†]	86.8 +10.0	76.8 +12.0	97.2 +2.1
LoRA, mode A	99.3 +22.5	98.2 +33.4	99.8 +4.7
LoRA, mode B	100.0 +23.2	100.0 +35.2	99.9 +4.8
Naive FT (full), mode A	99.9 +23.1	99.8 +35.0	99.7 +4.6
Naive FT (full), mode B	99.7 +22.9	99.4 +34.6	99.4 +4.3
Memory align (L=2, N=64)	99.8 +23.0	99.3 +34.5	99.7 +4.6

[†] GRPO reward = per-decision-step fraction picking f , gated on arithmetic validity (Run 6 of 7 explored variants; the best alignment-vs-arithmetic trade-off found). Binary chain-correctness reward (Run 1) collapsed pass@1 0.69→0.21; per-step ± 1 -on- f advantages (Run 7) lifted per-step f -rate to $\approx 76\%$ but broke arithmetic (blocks_valid 0.65, pass@1 0.31). No GRPO recipe achieved alignment and arithmetic simultaneously. GRPO evaluated at $n_{\text{eval}}=500$; all other rows at $n_{\text{eval}}=2000$.

Mode A vs. B

- **Mode A** — ffff-only data, GT labels (standard LM loss).
- **Mode B** — mixed data with f -override at decision points (the same alignment signal Memory uses).

Reading the results

1. **GRPO is the wrong tool for alignment.** Across 7 reward variants explored, none achieved alignment without breaking arithmetic. The shown row uses a dense per-step f -reward gated on arithmetic validity — the only variant that kept arithmetic intact, but moves f-selection only +7.7 pp and full-alignment +2.6 pp. More aggressive variants (per-step ± 1 advantages) reach $\sim 76\%$ per-step f -rate but collapse arithmetic (pass@1 $0.69 \rightarrow 0.31$). Every other aligning method exceeds +28 pp on f-selection while keeping op-acc.
2. **Memory alignment dominates the alignment $\times$ arithmetic trade-off.** Best f-selection (97.6 %) and best full-alignment (88.0%) on mixed val, with the smallest op-accuracy regression (-0.7 pp). And it does this with ~ 328 K trainable params — about 1% of the base model.
3. **Naive FT mode B exhibits catastrophic forgetting** (-10 pp op-accuracy). With 37.9M free parameters, the override-on-mixed-data signal warps arithmetic. LoRA at the same supervision stays stable because the adapter only moves the model in a low-rank subspace.
4. **LoRA mode A generalizes poorly off-distribution.** It saturates on ffff val (98.2% full align) but only reaches 37 % on mixed val — a 61 pp gap that exposes the limit of training on ffff-only data.